Homework 6

● Graded

Student

Colin Skinner

Total Points

100 / 100 pts

Question 1

Problem 1 — **12** / 12 pts

1.1 a — **3** / 3 pts

✔ **– 0 pts** Correct

1.2 b — **3** / 3 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect Proof

1.3 c — **3** / 3 pts

✔ **– 0 pts** Correct

1.4 d — **3** / 3 pts

✔ **– 0 pts** Correct

Question 2

Problem 2 — **20** / 20 pts

2.1 a — **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt Math Error

– 2 pts Flipped the signs of rotations

– 1 pt Didn't transpose

– 5 pts major error

2.2 b — **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

– 5 pts Incorrect angle of rotation

– 5 pts Incorrect eigenvector

– 10 pts No attempt

Question 3

Problem 3

18 / 18 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

Question 4

Problem 4

25 / 25 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

– 2 pts Math Error

– 5 pts Error in Rotation Sequences / Rotational Matrices

– 10 pts Does not get A

Question 5

Problem 5

25 / 25 pts

✔ **– 0 pts** Correct

Questions assigned to the following page: 1.1 and 1.2

$$\boldsymbol{A}=\begin{bmatrix}2&3&1\\4&1&6\\7&3&5\end{bmatrix},\boldsymbol{B}=\begin{bmatrix}6&3&3\\1&7&0\\1&2&8\end{bmatrix},\boldsymbol{C}=\begin{bmatrix}8&2&3\\6&3&9\\1&5&0\end{bmatrix}$$

Show that matrix multiplication is associative (i.e. $(\boldsymbol{AB})\boldsymbol{C}=\boldsymbol{A}(\boldsymbol{BC})$, meaning the order which multiplication is performed is not important).

$$AB=\begin{bmatrix}2&3&1\\4&1&6\\7&3&5\end{bmatrix}\begin{bmatrix}6&3&3\\1&7&0\\1&2&8\end{bmatrix}=\begin{bmatrix}16&29&19\\31&31&60\\50&52&61\end{bmatrix}$$

$$(AB)C=\begin{bmatrix}16&29&19\\31&31&60\\50&52&61\end{bmatrix}\begin{bmatrix}8&2&3\\6&3&9\\1&5&0\end{bmatrix}=\begin{bmatrix}316&189&309\\494&455&372\\773&561&618\end{bmatrix}$$

$$BC=\begin{bmatrix}6&3&3\\1&7&0\\1&2&8\end{bmatrix}\begin{bmatrix}8&2&3\\6&3&9\\1&5&0\end{bmatrix}=\begin{bmatrix}69&36&45\\50&23&66\\28&48&21\end{bmatrix}$$

$$A(BC)=\begin{bmatrix}2&3&1\\4&1&6\\7&3&5\end{bmatrix}\begin{bmatrix}69&36&45\\50&23&66\\28&48&21\end{bmatrix}=\begin{bmatrix}316&189&309\\494&455&372\\773&561&618\end{bmatrix}$$

b. Show that the trace of matrix $\boldsymbol{A}$ (given in part a) is unchanged under a similarity transformation with transformation matrix $\boldsymbol{B}$ (given in part a, you may also use mathematical properties of the trace).

$$\mathrm{tr}(A)=2+1+5=8$$

$$B^{-1}=\frac{1}{297}\begin{bmatrix}56&-18&-21\\-8&45&3\\-5&-7&39\end{bmatrix}\qquad BA=\begin{bmatrix}6&3&3\\1&7&0\\1&2&8\end{bmatrix}\begin{bmatrix}2&3&1\\4&1&6\\7&3&5\end{bmatrix}=\begin{bmatrix}45&30&39\\30&10&43\\66&29&53\end{bmatrix}$$

$$(BA)B^{-1}=\begin{bmatrix}45&30&39\\30&10&43\\66&29&53\end{bmatrix}\frac{1}{297}\begin{bmatrix}56&-18&-21\\-8&45&3\\-5&-7&39\end{bmatrix}=\frac{1}{297}\begin{bmatrix}2085&189&156\\1385&-477&1077\\3199&-360&768\end{bmatrix}$$

$$\mathrm{tr}(BAB^{-1})=\frac{1}{297}(2085-477+768)=\frac{2376}{297}=8$$

Question assigned to the following page: 1.3

$$A = \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{-1}{2} & 0 & \frac{\sqrt{3}}{2} \end{bmatrix}, B = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Show that the product is also orthogonal.

$$AB = \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{-1}{2} & 0 & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{\frac{3}{2}} & \sqrt{\frac{3}{2}} & \frac{1}{2} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$(AB)^T(AB) = \begin{bmatrix} \sqrt{\frac{3}{2}} & -\frac{1}{\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \sqrt{\frac{3}{2}} & \frac{1}{\sqrt{2}} & \frac{-1}{2\sqrt{2}} \\ \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \sqrt{\frac{3}{2}} & \sqrt{\frac{3}{2}} & \frac{1}{2} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question assigned to the following page: 1.4

$$\begin{bmatrix} 9 & -3 & 0 \end{bmatrix}$$

is preserved under the orthogonal similarity transformation with transformation matrix

$$\boldsymbol{B} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{2} \\ \frac{-1}{2} & \frac{\sqrt{3}}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$

Similarity transformation: $\boldsymbol{A'} = \boldsymbol{BAB}^{-1}$ where $\boldsymbol{B}$ is the transformation matrix, and where $\boldsymbol{A}, \boldsymbol{B} \in \boldsymbol{R}^{n \times n}$

$$BAB^{-1} = \begin{bmatrix} 0 & \frac{13\sqrt{3} - 3\sqrt{2}}{2\sqrt{2}} & \frac{-5}{\sqrt{2}} \\ \frac{-13\sqrt{3} + 3\sqrt{2}}{2\sqrt{2}} & 0 & \frac{13 + 3\sqrt{6}}{2\sqrt{2}} \\ \frac{5}{\sqrt{2}} & \frac{-6\sqrt{3} - 13\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\left(BAB^{-1}\right)^{T} = \begin{bmatrix} 0 & \frac{-13\sqrt{3} + 3\sqrt{2}}{2\sqrt{2}} & \frac{5}{\sqrt{2}} \\ \frac{13\sqrt{3} - 3\sqrt{2}}{2\sqrt{2}} & 0 & \frac{-6\sqrt{3} - 13\sqrt{2}}{4} \\ \frac{-5}{\sqrt{2}} & \frac{13 + 3\sqrt{6}}{2\sqrt{2}} & 0 \end{bmatrix}$$

Questions assigned to the following page: 2.1 and 2.2

a. Find the rotation matrix expressing its orientation with respect to its original frame (i.e. find $\boldsymbol{A}$ in $r = \boldsymbol{A}r'$ where r is the original frame)

b. If this was to be expressed as a rotation about an arbitrary axis, find the equivalent angle and axis.

a)

$$r' = R_{270} R_{\theta} r$$

$$r' = \begin{bmatrix} \cos 270^\circ & \sin 270^\circ & 0 \\ -\sin 270^\circ & \cos 270^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} r$$

$$r' = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} r$$

$$r' = \begin{bmatrix} 0 & -\cos\theta & -\sin\theta \\ 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} r \quad \rightarrow \quad r = \begin{bmatrix} 0 & -\cos\theta & -\sin\theta \\ 1 & 0 & 0 \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} r'$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -\cos\theta & 0 & -\sin\theta \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

b)

$$\mathrm{Tr}(A) = 1 + 2\cos\theta \qquad \phi = \cos^{-1}\left(\frac{\cos\theta - 1}{2}\right)$$

$$\phi = \cos^{-1}\left(\frac{\cos\theta - 1}{2}\right)$$

$$|R - \lambda I| = |R - I| = 0$$

$$\left| \begin{bmatrix} 0 & 1 & 0 \\ -\cos\theta & 0 & -\sin\theta \\ \sin\theta & 0 & \cos\theta \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| = 0 = \left| \begin{bmatrix} -1 & 1 & 0 \\ -\cos\theta & -1 & -\sin\theta \\ -\sin\theta & 0 & \cos\theta - 1 \end{bmatrix} \right|$$

Question assigned to the following page: 2.2

$-x + y$ ... y set x

$$-\sin\theta\, x + (\cos\theta - 1)z = 0 \qquad z = \frac{\sin\theta}{\cos\theta - 1}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \\ \frac{\sin\theta}{\cos\theta - 1} \end{bmatrix}$$

Question assigned to the following page: 3

$$A = \begin{bmatrix} \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & \frac{2}{4} \\ \frac{3}{4} - \frac{\sqrt{2}}{8} & \frac{(\sqrt{2}+2)\sqrt{3}}{8} & \frac{\sqrt{2}}{4} \\ -\frac{(\sqrt{2}+2)\sqrt{3}}{8} & \frac{3\sqrt{2}}{8} - \frac{1}{4} & \frac{\sqrt{6}}{4} \end{bmatrix}$$

Find the *Z-Y-X* Euler angles (ϕ, θ, ψ) for this rotation matrix (express your answer in degrees)

A_{13}

$-\sin\theta = \frac{\sqrt{2}}{2}$

$\theta = -45°$ on y axis

ϕ = x rot θ = Y rot ψ = Z rot

A_{12}

$\sin\psi\cos\theta = \sin\psi\cos(-45°) = \frac{\sqrt{2}}{2}\sin\psi = -\frac{\sqrt{6}}{4} \rightarrow \sin\psi = -\frac{\sqrt{3}}{2}$

$\psi = -60°$

$\psi = 300°$ on z axis

A_{23}

$\cos\theta\sin\phi = \frac{\sqrt{2}}{4} = \cos(-45°)\sin\phi = \frac{\sqrt{2}}{2}\sin\phi = \frac{\sqrt{2}}{4}$ $\sin\phi = \frac{1}{2}$

$\phi = 30°$ on x axis

Question assigned to the following page: 4

angular rate $(\omega_x, \omega_y, \omega_z)$:

$$\begin{bmatrix}\dot\phi\\ \dot\theta\\ \dot\psi\end{bmatrix} = [A]\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix}$$

$\dot{\vec\theta}$ $\vec\omega$

ϕ = x rot

θ = y rot

ψ = z rot

$$\vec\omega = \dot\phi\hat z + \dot\theta\hat y' + \dot\psi\hat x''$$

Solve for A.

r''

$$\hat\omega_{\phi r} = \overleftrightarrow{\phi}_x \hat x'' = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin & \cos\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$\vec\omega_{\theta r} = \overleftrightarrow{\phi}_x\overleftrightarrow{\theta}_y \hat y' = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\ 0 & 1 & 0\\ \sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\\cos\phi\\-\sin\phi\end{bmatrix}$$

$$\vec\omega_{\psi r} = \overleftrightarrow{\phi}_x\overleftrightarrow{\theta}_y\overleftrightarrow{\psi}_z\hat z = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\ 0 & 1 & 0\\ \sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\ 0 & 1 & 0\\ \sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\phi & \sin\phi\\ 0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}-\sin\theta\\0\\\cos\theta\end{bmatrix}$$

$$= \begin{bmatrix}-\sin\theta\\ \cos\theta\sin\phi\\ \cos\theta\cos\phi\end{bmatrix}$$

$$\vec\omega_r = \overleftrightarrow{C}\begin{bmatrix}\dot\phi\\\dot\theta\\\dot\psi\end{bmatrix} = \begin{bmatrix}1 & 0 & -\sin\theta\\ 0 & \cos\phi & \cos\theta\sin\phi\\ 0 & -\sin\phi & \cos\theta\cos\phi\end{bmatrix}$$

$$\overleftrightarrow{A} = \overleftrightarrow{C}^{-1} = \frac{1}{\det(C)}\,adj(C)^T$$

$$\det(C) = \cos\theta\cos^2\phi + \cos\theta\sin^2\phi = \cos\theta$$

Question assigned to the following page: 4

$$Adj(A) = Adj\begin{pmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & c\theta s\phi \\ 0 & -s\phi & c\theta c\phi \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \cos\theta\sin\phi \\ 0 & -\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

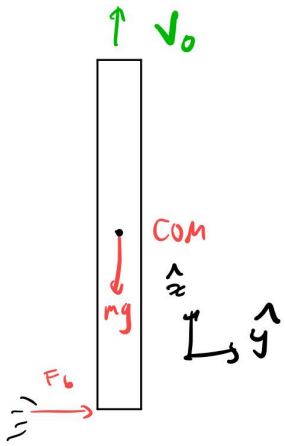
$$\begin{bmatrix} \begin{vmatrix} c\phi & c\theta s\phi \\ -s\phi & c\theta c\phi \end{vmatrix} & -\begin{vmatrix} 0 & -s\theta \\ -s\phi & c\theta c\phi \end{vmatrix} & \begin{vmatrix} 0 & -s\theta \\ c\phi & c\theta s\phi \end{vmatrix} \\ -\begin{vmatrix} 0 & c\theta s\phi \\ 0 & c\theta c\phi \end{vmatrix} & \begin{vmatrix} 1 & -s\theta \\ 0 & c\theta c\phi \end{vmatrix} & -\begin{vmatrix} 1 & -s\theta \\ 0 & c\theta s\phi \end{vmatrix} \\ \begin{vmatrix} 0 & c\phi \\ 0 & -s\phi \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 0 & -s\phi \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 0 & c\phi \end{vmatrix} \end{bmatrix} = \begin{bmatrix} c\theta & s\theta s\phi & s\theta c\phi \\ 0 & c\theta c\phi & -c\theta s\phi \\ 0 & s\phi & c\phi \end{bmatrix}$$

$-c\theta s^2\phi - c\theta c^2\phi$

$$\overleftrightarrow{A} = \frac{1}{\cos\theta}\begin{pmatrix} c\theta & s\theta s\phi & s\theta c\phi \\ 0 & c\theta c\phi & -c\theta s\phi \\ 0 & s\phi & c\phi \end{pmatrix} = \begin{bmatrix} 1 & \tan\theta\sin\phi & \tan\theta\cos\phi \\ 0 & \cos\phi & -\sin\phi \\ 0 & \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} \end{bmatrix}$$

Question assigned to the following page: 5

...ou will ...ics and ...ext few homeworks (homework 6-8), and it will become your final project for the course. For this week, describe in words and draw a picture of a dynamics problem (e.g. aircraft, spacecraft, space station). Don't feel limited to real aircraft or spacecraft; use sci-fi as inspiration if you wish. The problem will be a bit more complicated than your first problem and must be in 3D and involve rigid body dynamics (i.e. must use moments of inertia in your solution). What are you trying to solve? What is included in your system? Choose your reference frame and coordinate system and label it on your drawing. Explain why you chose what you did. Identify and label all of the relevant forces (i.e. create a free-body diagram, aka FBD). Do NOT go beyond this – this week is meant for you to carefully describe your problem visually and in words. Be creative and specific with your setup (do not just say "I am going to solve the EOM for an aircraft".)



(Not neglecting gravity)

$\vec{r}_0 = [x_0, y_0, z_0]^T$

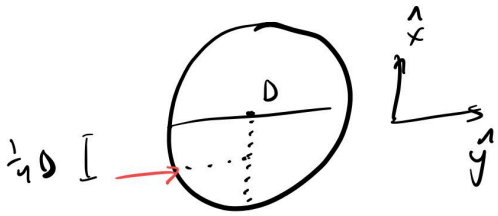
$\vec{v}_0 = [0, 0, V_0]^T$

$x(0) = x_0 \quad y(0) = y_0$

$\dot{x}(0) = 0 \quad \dot{y}(0) = V_0$

$\omega(0) = 0 \quad \alpha(0) = 0$

with rocket mass M diameter D length L



what is the maximum altitude the rocket will reach, and how far from its launch site will it land?

Reference/Coord Frames

- Body (Cartesian)
- Inertial (Cartesian)

Nothing is conserved